

Using GitHub with VS Code

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What is GitHub?

- git: version control system
 - retaining software history
 - testing different ideas
- GitHub: online collaborative storage for git
 - keeps a backup of your project and history in repo (repository)
 - backing up absolutely necessary for any larger project (IA)
 - allows you to share and co-operate
 - free version allows both public and private repos

GitHub prerequisites

- create account at <https://github.com>
 - to help AI adoption later
 - use school email (@syk.fi) as email address
 - enable 2FA (two-factor authentication)
- Windows: install git via Company Portal / Yritysportaali

General GitHub workflow (single-author project)

1. create local repo in your computer
2. create remote repo on GitHub
3. link local and remote
4. repeat
 - 4.1 make changes locally by editing code
 - 4.2 stage changes for the next commit using git
 - 4.3 commit changes locally using git
 - 4.4 push changes to remote using git

Put project under source control (once per project)

1. open project (folder) or create new one
2. initialize local repo (ctrl-shift-p *Git Initialize Repository*)
3. commit file locally
 - 3.1 open source control panel
 - 3.2 stage changes (+)
 - 3.3 commit
 - 3.3.1 write commit message
 - 3.3.2 save (ctrl-s) and close (ctrl-w)
4. *Publish Branch* (links to remote and pushes)

Commit and push (once per significant change)

1. make any changes into your project, save files
2. commit file locally
 - 2.1 open source control panel
 - 2.2 stage changes (+)
 - 2.3 write commit message and commit
3. *Sync Changes* (pushes)

Programmers think to avoid writing the same thing again
A good programmer is like:

- there are many reasons for this
 - saving time: how you are writing it the first time
 - (often) shorter code
 - easier to understand: "some thing happens here"
 - easier to maintain: only change one function
 - easier to debug: isolates problem to one place
 - test is abstraction: patterns emerge
- if you are repeating yourself, think again and generalize
- functions are a fundamental method of generalization
- big code written only once is often split into procedures
 - structuring
 - division of labor